

 Unison Laboratories Co., Ltd.	APPROVED PACKAGING MATERIAL (EXPORT)	☒ P.O. No. 007 / 2008
		☒ Endorsement ☒ New ☐ Revision
Product Name (Local) : TOLBIN Expectorant Syrup	Reg. No. (Local) : 2A 6/45	
Product Name (Export) : TOLBIN Expectorant Syrup	Reg. No. (Export) : MAL07090910A	
Active Name : Terbutaline sulfate/ Guaifenesin	Reg. Approval (Export) :	
Customer : MALAYSIA (Medispec)	Reg. Validity (Export) :	

Tolbin *Expectorant* Syrup

BRONCHODILATOR/ EXPECTORANT

Each teaspoonful (5 mL) of syrup contains:
Terbutaline sulfate...1.5 mg
Glyceryl guaiacolate (Guaifenesin)...66.5 mg

● PRODUCT DESCRIPTION:

Clear, colorless syrup with menthol sweet orange-soda flavor

● MECHANISM OF ACTION:

Pharmacology

Terbutaline: Sympathomimetic agents mimic, to varying degrees, the actions of endogenous catecholamines by stimulating adrenergic receptors. Stimulation of adrenergic receptors may occur directly (e.g., epinephrine) or indirectly via norepinephrine release (e.g., ephedrine), or effects may be mixed. Relative stimulation of alpha- and beta-adrenergic receptors and selectivity of effect (alpha₁, alpha₂, beta₁, beta₂) vary with individual agents, which leads to variation in effects.

Primarily stimulates beta₂-adrenergic receptors, with some minor beta₁-adrenergic activity.

Bronchodilator: Adrenergic bronchodilators act by stimulating beta₂-adrenergic receptors in the lungs to relax bronchial smooth muscle, thereby relieving bronchospasm.

Glyceryl guaiacolate (Guaifenesin): Guaifenesin is thought to act as an expectorant by increasing the volume and reducing the viscosity of secretions in the trachea and bronchi. Thus, it may increase the efficiency of the cough reflex and facilitate removal of the secretions; however, objective evidence for this is limited and conflicting.

Pharmacokinetics

Terbutaline:

Absorption: Oral: 30 to 70%; food reduces bioavailability by one third.

Biotransformation: Gastrointestinal and hepatic, 60% metabolized by first-pass metabolism; no known active metabolites.

Half-life: Elimination - 3 to 4 hours.

Onset of action: Measurable change in flow rate - Within 30 minutes. Improvement in pulmonary function - Within 60 to 120 minutes.

Time to peak concentration: Serum - After single dose: 30 minutes to 5 hours.

Peak plasma concentration: Average of approximately 1 nanogram per mL after each 1 mg administered to fasting adults.

Time to peak effect: 120 to 180 minutes.

Duration of action: Clinically significant decrease in airway and pulmonary resistance - At least 4 hours.

Significant bronchodilator activity, as measured by various pulmonary function determinations (airway resistance, MMEFR, PEFr) - Up to 8 hours.

Elimination: Renal. In dialysis - It is not known whether terbutaline is removable by dialysis.

Glyceryl guaiacolate (Guaifenesin):

Absorption: Readily absorbed from gastrointestinal tract

Elimination: Renal, as inactive metabolites

● INDICATION:

It is indicated in conditions where both bronchodilation and expectoration effects are required such as in chronic bronchitis and emphysema.

● DOSAGE AND ADMINISTRATION:

Oral

Usual adult dose: 2 to 3 teaspoonful

Usual pediatric dose: Children 7 to 15 years - 1 to 2 teaspoonful

Two to three times daily or as prescribed by physician.

● CONTRAINDICATION:

Contraindicated in patients with a history of hypersensitivity to any of their constituents.

● PRECAUTION:

Terbutaline sulfate:

Care should be taken with patients suffering from myocardial insufficiency or thyrotoxicosis.

Due to the hyperglycaemic effects of β₂-stimulants, additional blood glucose measurements are initially recommended when TOLBIN Expectorant therapy is commenced in diabetic patients.

If a previously effective dosage regimen no longer gives the same symptomatic relief, the patient should urgently seek further medical advice.

Consideration should be given to the requirements for additional therapy (including increased dosages of anti-inflammatory medication).

Severe exacerbation of asthma should be treated as an emergency in the usual manner.

Potentially serious hypokalaemia may result from β₂-agonist therapy, mainly with parenteral or nebulised administration.

Particular caution is advised in acute severe asthma as this effect may be augmented by hypoxia.

The hypokalaemic effect may be potentiated by concomitant treatment with xanthine derivatives, corticosteroids and/ or diuretics.

It is recommended that serum potassium levels be monitored in such situations.

Due to positive inotropic effect of β₂-agonists, these drugs should not be used in patients with hypertrophic cardiomyopathy.

Glyceryl guaiacolate (Guaifenesin):

Urinary calculi have been reported in patients consuming large quantities of over-the-counter preparations containing Glyceryl guaiacolate (Guaifenesin).

Glyceryl guaiacolate (Guaifenesin) is considered to be unsafe in patients with porphyria because it has been shown to be porphyrinogenic in animals.

Pregnancy/ Lactation

Although no teratogenic effects have been observed in animals or in patients. TOLBIN Expectorant should only be administered with caution during the first trimester of pregnancy. Terbutaline is secreted in breast milk, but effect on the infant is unlikely at therapeutic doses.

Product Apper

PM Descriptio

PM Code_Rev

PM Approval

Prepared by:

● **DRUG INTERACTION:**

Beta-blocking agents (including eye drops), especially the non-selective ones such as Propanolol may partially or totally inhibit the effect of β -stimulants. Therefore, TOLBIN Expectorant preparations and non-selective β -blockers should not normally be administered concurrently.

TOLBIN Expectorant should be used with caution in patients receiving other sympathomimetics. Hypokalemia may result from β_2 -agonist therapy and may be potentiated by concomitant treatment with xanthine derivatives, corticosteroids and diuretics.

● **SIDE EFFECT:**

The frequency of side effects is low at the recommended doses.

Side effects which have been recorded such as tremor, headache, nausea, mouth and throat irritation, tonic cramp and palpitations are all characteristic of sympathomimetic amines.

A few patients feel tense, this is also due to the effects on skeletal muscle and not to direct CNS stimulation.

Whenever these side effects have occurred the majority have usually been spontaneously reversible within the first week of treatment. As with other β_2 -agonists, tremor is dose related.

Sleep disturbances and behavioral disturbances, such as agitation, hyperactivity and restlessness have been observed.

Tachycardia, with or without peripheral vasodilation, has been reported during β_2 -agonist therapy. Cardiac arrhythmia, including atrial fibrillation, supraventricular tachycardia and extrasystoles have been reported in association with β_2 -agonists, usually in susceptible patients.

Potentially serious hypokalaemia may result from β_2 -agonist therapy.

TOLBIN Expectorant therapy should be discontinued and after assessment, an alternative therapy initiated.

Hypersensitivity reactions, including angioedema, urticaria, exanthema, bronchospasm, hypotension and collapse, have been very rarely reported with β_2 -agonist therapy.

Glyceryl guaiacolate (Guaifenesin):

Gastrointestinal discomfort, nausea, and vomiting have occasionally been reported with Glyceryl guaiacolate (Guaifenesin), particularly in very large doses.

● **OVERDOSE AND TREATMENT:**

Possible symptoms and signs

Headache, anxiety, tremor, nausea, tonic cramp, palpitations, tachycardia, arrhythmia. A fall in blood pressure sometimes occurs.

Laboratory findings: Hypokalaemia, hyperglycaemia and lactic acidosis sometimes occur.

Treatment

Mild and moderate cases: Reduce the dose.

Severe cases:

Gastric lavage, administration of activated charcoal.

Determination of acid-base balance, blood sugar and electrolytes, particularly serum potassium levels.

Monitoring of the heart rate and rhythm and blood pressure.

Metabolic changes should be corrected.

A cardioselective β -blocker (e.g. Metoprolol) is recommended for the treatment of arrhythmias causing haemodynamic deterioration. The β -blocker should be used with care because of the possibility of inducing bronchoconstriction. Use with caution in patients with a history of bronchospasm.

If the β_2 -mediated reduction in the peripheral vascular resistance significantly contributes to the fall in blood pressure, a volume expander should be given.

● **STORAGE:**

Store at temperature of not more than 30°C.

● **SHELF LIFE:**

Three (3) years

● **SUPPLY:**

PET Bottle by 100 mL

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